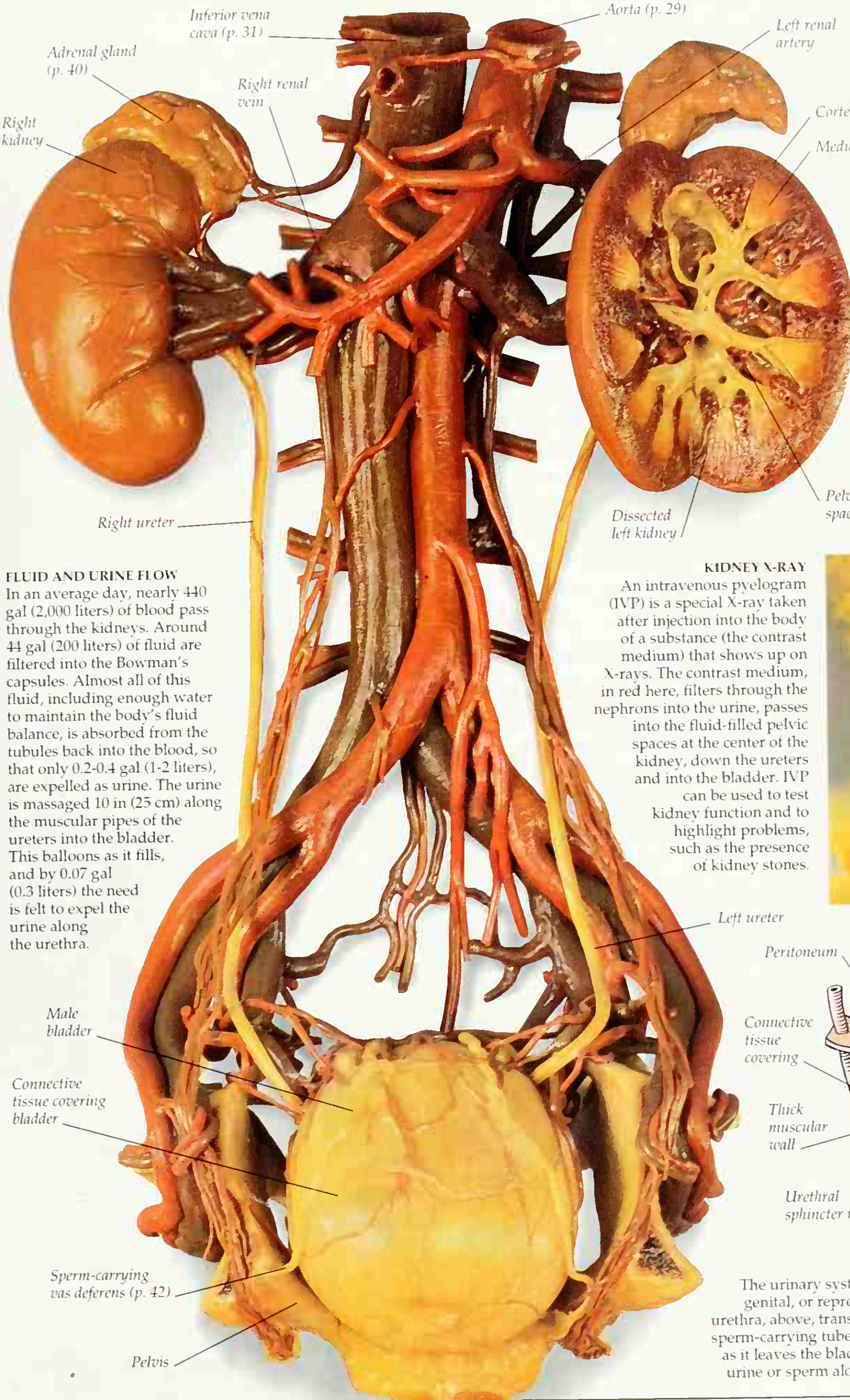
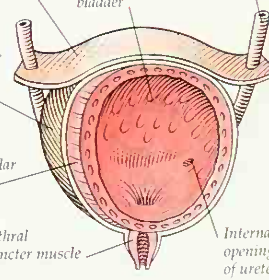


ONE IN THREE MILLION
Sweat is cooling – and excreting. About 3 million microscopic sweat glands (one is magnified here) are concentrated on the hands and feet, forehead, armpits, and pubic region. Sweat oozes gently all the time, and runs copiously when the body is too hot. The water evaporates, cooling the body. The substances in sweat, including body salts, and wastes like urea, are left to dry on the skin.

FLUID AND URINE FLOW
In an average day, nearly 440 gal (2,000 liters) of blood pass through the kidneys. Around 44 gal (200 liters) of fluid are filtered into the Bowman's capsules. Almost all of this fluid, including enough water to maintain the body's fluid balance, is absorbed from the tubules back into the blood, so that only 0.2-0.4 gal (1-2 liters), are expelled as urine. The urine is massaged 10 in (25 cm) along the muscular pipes of the ureters into the bladder. This balloons as it fills, and by 0.07 gal (0.3 liters) the need is felt to expel the urine along the urethra.

KIDNEY X-RAY
An intravenous pyelogram (IVP) is a special X-ray taken after injection into the body of a substance (the contrast medium) that shows up on X-rays. The contrast medium, in red here, filters through the nephrons into the urine, passes into the fluid-filled pelvic spaces at the center of the kidney, down the ureters and into the bladder. IVP can be used to test kidney function and to highlight problems, such as the presence of kidney stones.



FEMALE URINARY SYSTEM
The urinary system has close connections with the genital, or reproductive, system. The short female urethra, above, transports only urine. In the male (left) sperm-carrying tubes join the urine-conveying urethra as it leaves the bladder, so the urethra conveys either urine or sperm along the penis to the outside (p. 42).